Outcome Analysis Validation Report

Model Name

Mortgage Insurance Credit Risk Model

Validation Objective

The purpose of this review is to assess whether the model outputs are accurate, reliable, stable, appropriately calibrated, and suitable for the intended mortgage insurance credit risk use case.

The review focused on:

- Predictive performance
- Calibration and reliability
- Segment-level behavior
- Stability across vintages and regions
- Sensitivity analysis
- Stress testing behavior
- Back-testing results
- Bias and fairness considerations
- Overfitting risk

The assessment was performed using the implemented modeling, preprocessing, and feature engineering framework.

---

1. Error Diagnostics and Performance Decomposition

Objective

To assess whether model errors are measured appropriately and whether performance differences across segments are understood.

Description

The review evaluated overall predictive performance and assessed whether model errors vary materially across products, regions, borrower segments, and time periods.

Validation Assessment

Review Item	Status
Overall model error quantified	Yes
Error reviewed across segments	Partial
Systematic underprediction or overprediction identified	Partial
Error patterns explained in business terms	Partial
Segment-level performance documented	Partial

Validator Comments

The framework supports segmentation by:

- Region
- Economic segment
- Borrower characteristics
- Vintage
- LOB classification

The validator observed that the structure supports performance decomposition analysis; however, formal segment-level error reporting and business interpretation should be expanded.

Additional analysis should assess whether performance deteriorates in specific borrower or regional segments.

---

2. Prediction Reliability

Objective

To assess whether model predictions are reliable, calibrated, and appropriate across the full range of risk.

Description

The review evaluated whether predicted outcomes align reasonably with observed outcomes and whether the model behaves consistently across different risk levels.

Validation Assessment

Review Item	Status
Predictions compared against observed outcomes	Yes
Calibration assessed across score bands	Partial
Predictions reasonable across risk spectrum	Yes
Errors not concentrated in subgroups	Partial
Reliability aligned with intended use	Yes

Validator Comments

The model framework appears capable of producing economically reasonable predictions across borrower and collateral risk levels.

However, additional calibration analysis by:

- Risk band
- Vintage
- Region
- Product segment

would improve confidence in prediction reliability and portfolio-level consistency.

---

3. Uncertainty Estimation

Objective

To assess whether model uncertainty is appropriately estimated and communicated.

Description

The review evaluated whether prediction uncertainty and model confidence are understood, particularly in areas with limited data or stressed economic conditions.

Validation Assessment

Review Item	Status
Prediction uncertainty estimated	No
Confidence intervals reasonable	No
Uncertainty estimates calibrated	No
High-uncertainty areas identified	Partial
Excessive confidence avoided	Partial

Validator Comments

No formal uncertainty estimation framework was identified during review.

While the model incorporates segmentation and economic variables that may indirectly capture uncertainty, explicit confidence intervals or prediction uncertainty measures were not observed.

Additional uncertainty analysis is recommended, particularly for stressed or low-volume portfolio segments.

---

4. Bias and Fairness Analysis

Objective

To assess whether the model exhibits unintended bias or materially different behavior across relevant portfolio segments.

Description

The review evaluated whether prediction quality and model behavior remain reasonably consistent across borrower groups, regions, and portfolio segments.

Validation Assessment

Review Item	Status
Bias assessed across relevant segments	Partial
Performance differences across groups reviewed	Partial
Proxy variables considered for fairness impact	Partial

Review Item	Status
No unexplained discriminatory patterns identified	Partial
Fairness concerns documented appropriately	Partial

Validator Comments

The framework includes:

- Regional segmentation
- Borrower-level aggregation
- Economic variables
- Credit bureau variables

No explicit fairness-testing framework was identified during review. Additional bias analysis is recommended to assess:

- Regional impacts
- Segment-level performance differences
- Potential proxy effects within engineered variables

---

5. Stability Analysis

Objective

To assess whether model performance and output distributions remain stable over time and across segments.

Description

The review evaluated whether score distributions and model behavior remain reasonably consistent across vintages, economic conditions, and borrower groups.

Validation Assessment

Review Item	Status
Performance tested over time	Partial
Score distributions stable across vintages	Partial

Review Item	Status
Segment behavior stable	Partial
Material drift identified and explained	Partial
Stability concerns assessed for impact	Partial

Validator Comments

The framework includes multiple years of data and economic-region segmentation, which supports stability analysis.

However, formal monitoring procedures such as:

- Population Stability Index (PSI)
- Characteristic Stability Index (CSI)
- Drift monitoring

were not identified during review.

Additional stability monitoring procedures are recommended.

---

6. Sensitivity Analysis

Objective

To assess whether model outputs respond appropriately to changes in key risk drivers.

Description

The review evaluated whether changes in borrower risk characteristics, collateral quality, and economic conditions produce economically reasonable changes in model outputs.

Validation Assessment

Review Item	Status
Key variables varied for testing	Partial
Output changes directionally sensible	Yes
Model not overly sensitive to small changes	Partial

Review Item	Status
Sensitivity aligned with risk intuition	Yes
Key output drivers identified	Partial

Validator Comments

The framework incorporates several economically intuitive drivers, including:

- Bureau scores
- Loan-to-income ratios
- Property values
- Economic variables

The validator expects worsening borrower affordability and economic conditions to increase mortgage credit risk, which appears directionally consistent with the implemented framework.

Formal sensitivity-testing documentation should be expanded.

---

7. Stress Testing

Objective

To assess whether the model behaves reasonably under adverse economic or portfolio conditions.

Description

The review evaluated whether the framework can support stressed economic scenarios and whether model outputs remain interpretable during adverse conditions.

Validation Assessment

Review Item	Status
Stress scenarios documented	Partial
Model tested under adverse conditions	Partial
Stress results plausible	Partial

Review Item	Status
Model usable under severe conditions	Partial
Stress weaknesses documented	Partial

Validator Comments

The framework incorporates macroeconomic variables such as:

- House Price Index (HPI)
- Unemployment rate (UE)

These variables support stress testing and economic sensitivity analysis. However, no formal stress-testing framework or documented adverse scenarios were identified during review.

Additional stress-testing procedures are recommended.

---

8. Back-Testing

Objective

To assess whether predicted outcomes align with realized outcomes over time.

Description

The review evaluated whether historical model performance has been compared against observed portfolio behavior across different segments and vintages.

Validation Assessment

Review Item	Status
Predicted outcomes compared with actual outcomes	Partial
Back-testing performed by segment and vintage	Partial
Forecast deviations analyzed	Partial
Back-testing methodology appropriate	Partial
Results used for remediation assessment	Partial

Validator Comments

The framework structure supports historical comparison analysis through:

- Vintage segmentation
- Regional segmentation
- Policy status tracking

However, formal back-testing documentation and remediation thresholds were not identified during review.

Additional back-testing procedures should be implemented before production deployment.

---

9. Overfitting Assessment

Objective

To assess whether model complexity is appropriate and whether the framework generalizes reasonably out of sample.

Description

The review evaluated whether the preprocessing, feature engineering, and modeling complexity may introduce overfitting risk.

Validation Assessment

Review Item	Status
Model complexity appropriate	Yes
Training and validation performance compared	Partial
No material out-of-sample deterioration identified	Partial
Complex features reviewed	Yes
Overfitting risk documented and mitigated	Partial

Validator Comments

The framework includes:

- Borrower aggregation

- Economic features
- Ratio features
- Regional segmentation

The overall feature complexity appears reasonable for mortgage insurance credit risk modeling.

However, additional out-of-sample testing and stability analysis would strengthen confidence that the model generalizes appropriately.

---

10. Overall Conclusion

Validation Assessment

Conclusion Item	**Status**
Outcome analysis acceptable	Yes
Outcome analysis acceptable subject to remediation	Yes
Outcome analysis not acceptable	No

Conclusion

The outcome analysis indicates that the model framework is generally capable of producing reasonable and economically intuitive results for mortgage insurance credit risk assessment.

Key strengths include:

- Economically meaningful risk relationships
- Integration of borrower and macroeconomic variables
- Structured segmentation framework
- Appropriate underwriting focus
- Reasonable feature complexity

The following improvements are recommended:

- Expand calibration analysis
- Implement formal stability monitoring

- Add uncertainty estimation procedures
- Improve stress-testing documentation
- Expand back-testing and sensitivity analysis

Overall, the validator considers the outcome analysis acceptable subject to remediation of identified monitoring, calibration, and governance gaps.

Key Findings

- The framework supports reasonable mortgage credit risk assessment.
- Risk relationships appear economically intuitive.
- Segment-level monitoring and calibration analysis should be expanded.
- Formal uncertainty estimation was not identified.
- Additional stress testing and back-testing procedures are recommended.

Required Actions

Action	Priority	Owner
Implement calibration monitoring by segment	Moderate	Model Development
Add formal stability monitoring procedures	Moderate	Model Risk
Develop uncertainty estimation framework	Moderate	Validation Team
Expand stress-testing scenarios	High	Model Risk
Implement formal back-testing procedures	High	Model Development
Perform additional out-of-sample testing	Moderate	Validation Team

---

Suggested Validation Wording

"Outcome analysis was performed to assess model performance, prediction reliability, uncertainty estimation, bias, stability, sensitivity, stress behavior, back-testing results, and overfitting risk. The review indicates that the model is generally performing as expected for its intended mortgage insurance use, subject to identified calibration, monitoring, and governance improvements."